pfi

Python for Fixed Income

COURSE 10 · IMPLEMENTING RAG

BM25 and Hybrid Search

WEEK 2 · DAY 1

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Today: BM25 and hybrid search

Three ways to rank one 10-K

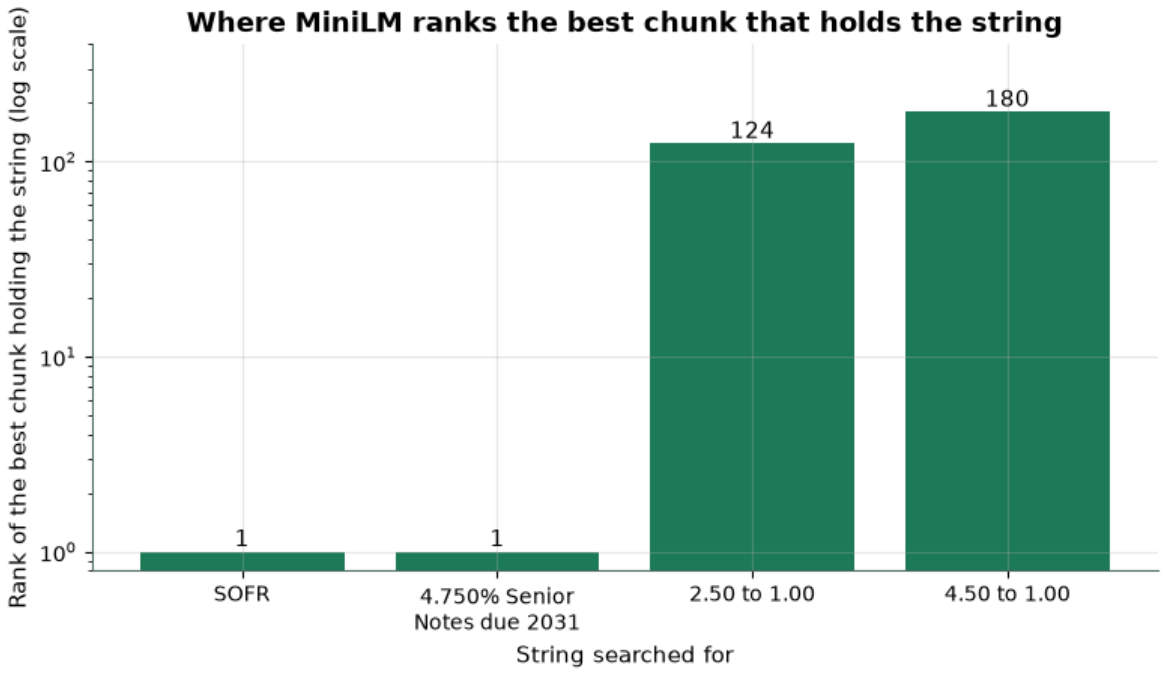
Chunks	1,158 at 512/64
MiniLM	ranks by meaning (day 4)
BM25	ranks by shared words
Hybrid	fuses the two lists
Model	none today; nothing is sent

Today

1. Keyword search is not dead
2. BM25 by hand
3. BM25 on the 10-K; the library
4. Counting hits on 23 questions
5. Reciprocal rank fusion
6. Hybrid against each list alone

Count on development questions; name the rule; choose nothing yet.

Keyword search is not dead



Chunks holding the string

SOFR

17 chunks; best rank 1 of 1,158; 10 in the top 10

4.750% Senior Notes due 2031

14 chunks; best rank 1 of 1,158; 4 in the top 10

2.50 to 1.00

3 chunks; best rank 124 of 1,158; 0 in the top 10

4.50 to 1.00

3 chunks; best rank 180 of 1,158; 0 in the top 10

arc04's page 38 chunk with 4.50 to 1.00: MiniLM ranks it 54 of 1,158 for the question as worded.

BM25 by hand: senior notes 2031

Three parts

```
idf = ln((N - n + 0.5) / (n + 0.5) + 1)

count = f (k1 + 1) / (f + k1 (1 - b + b dl / avgdl))

weight = idf x count; score = sum of weights

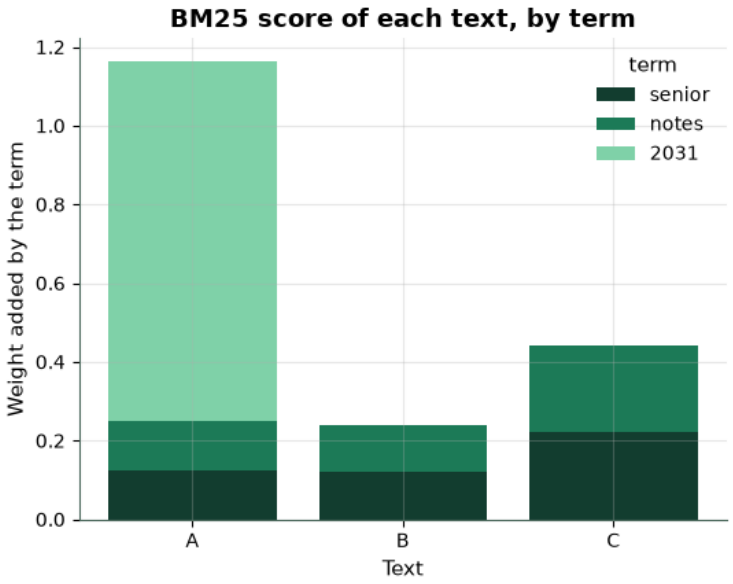
k1 = 1.5, b = 0.75; length in tokens
```

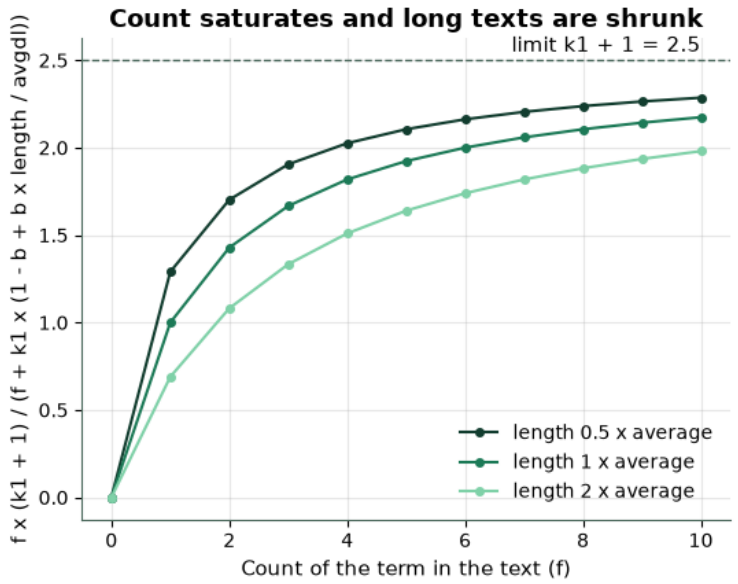
scores: A 1.1634, B 0.2393, C 0.4410

text	term	f	n	idf	dl	weight
A	senior	1	3	0.1335	12	0.1245
B	senior	1	3	0.1335	13	0.1196
C	senior	2	3	0.1335	6	0.2205
A	notes	1	3	0.1335	12	0.1245
B	notes	1	3	0.1335	13	0.1196
C	notes	2	3	0.1335	6	0.2205
A	2031	1	1	0.9808	12	0.9145
B	2031	0	1	0.9808	13	0.0000
C	2031	0	1	0.9808	6	0.0000

2031 is in one text, so it does the work; repeats and length are discounted.

Terms add up, counts level off





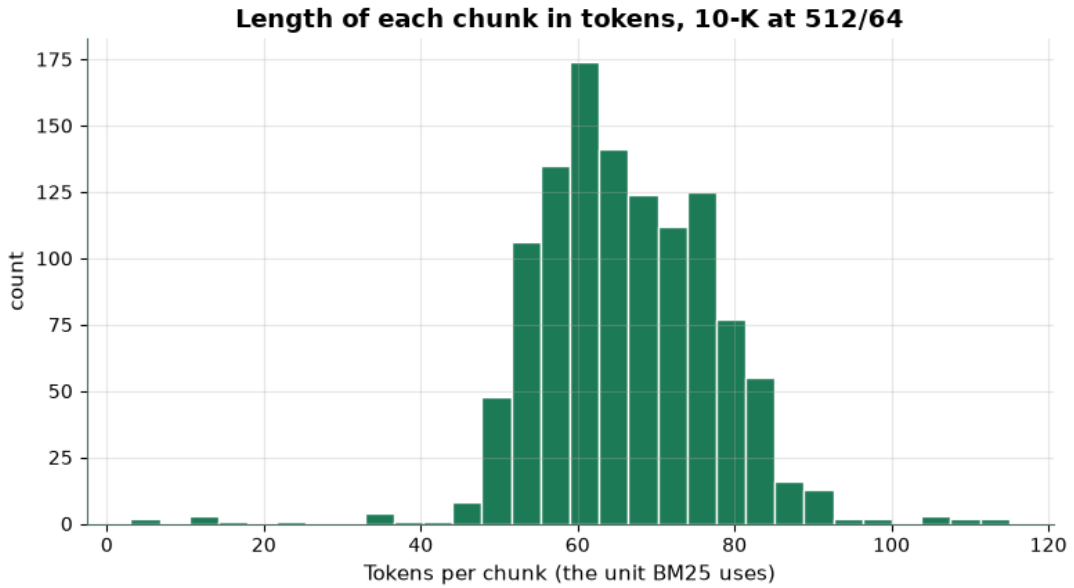
Where BM25 comes from

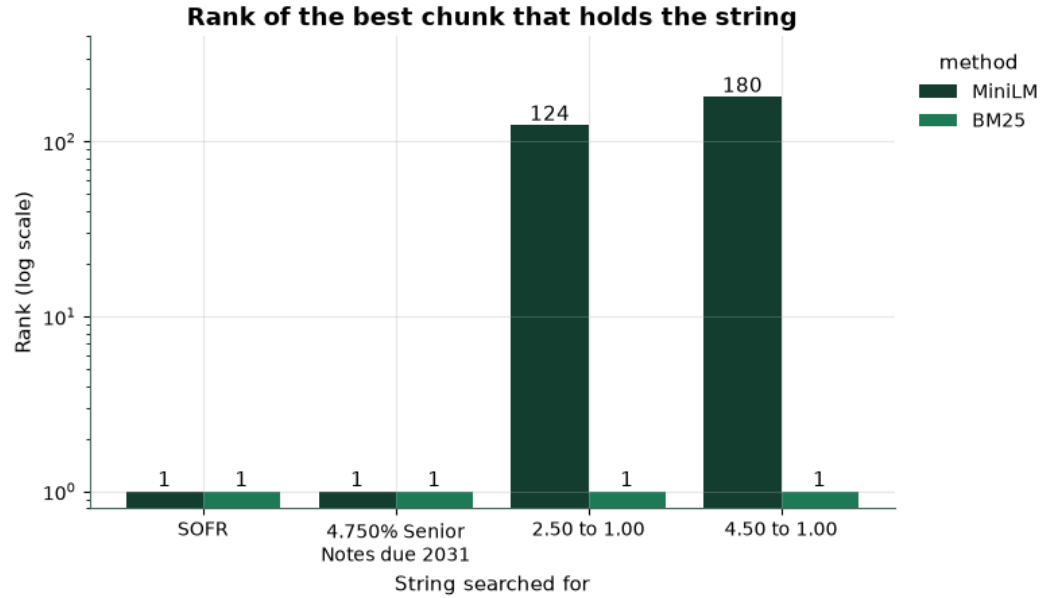
Robertson and Zaragoza, The Probabilistic Relevance Framework: BM25 and Beyond. Foundations and Trends in Information Retrieval 4(1-2), pp. 1-174, published 17 September 2009.

Crossref's record of DOI 10.1561/1500000019, read 13 October 2026. The paper itself was not read.

A term repeated ten times is not ten times a term seen once; a long text is shrunk.

BM25 on the 10-K





3-115

tokens per chunk

66.06

mean (avgdl)

426.8

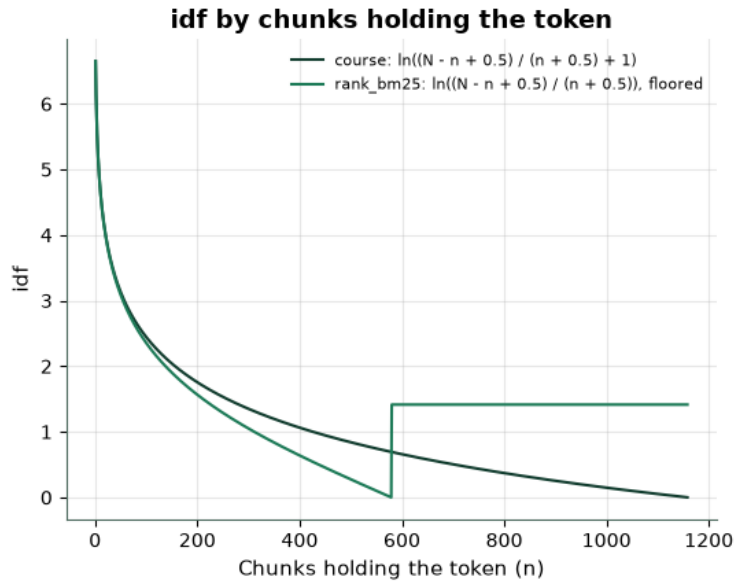
characters, mean

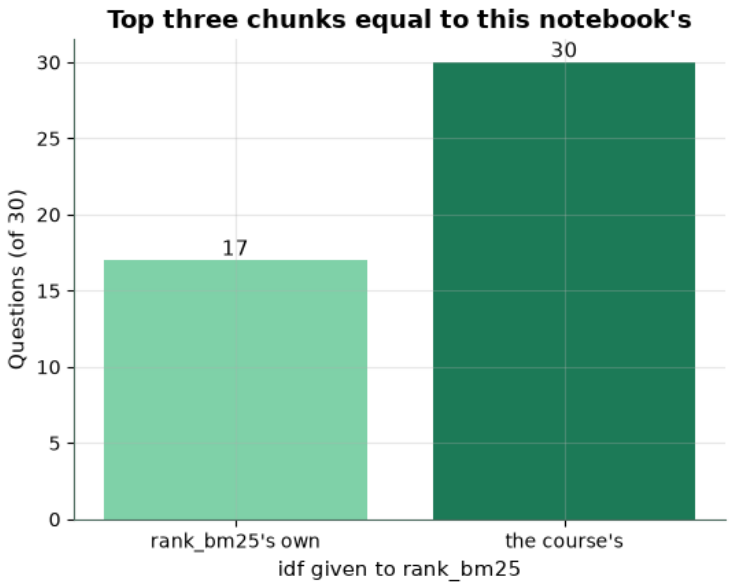
975

chunks scoring 0

Length in tokens throughout; BM25 ranks a chunk with each covenant level first, MiniLM does not.

How the rank_bm25 library differs





30 questions, top 3 chunks

own idf: 17 of 30 equal

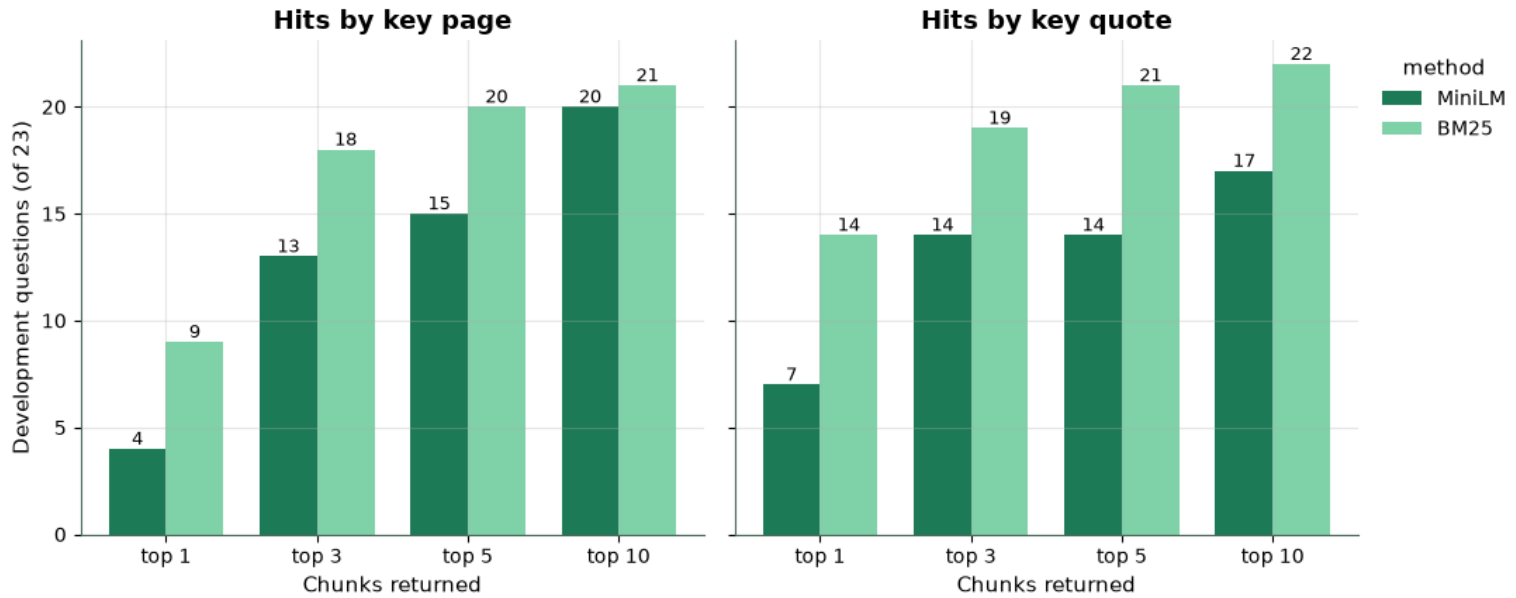
largest score gap 12.9

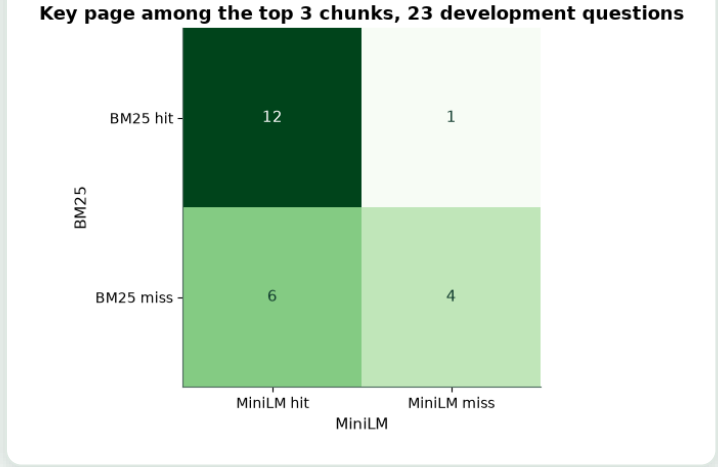
course idf put in: 30 of 30

scores equal within 1e-12

token	chunks	course idf	rank_bm25 idf					
the	1008	0.1391	1.4143		4.750	42	3.3058	3.2684
of	1023	0.1243	1.4143		sofr	17	4.1931	4.1779
senior	78	2.6922	2.6221					

MiniLM and BM25 on 23 development questions





method	rule	top 1	top 3	top 5	top 10
MiniLM	key page	4	13	15	20
MiniLM	key quote	7	14	14	17
BM25	key page	9	18	20	21
BM25	key quote	14	19	21	22

Page rule and quote rule, named each time; a description of these questions, not a ranking of methods.

Reciprocal rank fusion

A place in a list is worth 1 / (c + place)

Sum over the lists a chunk is in

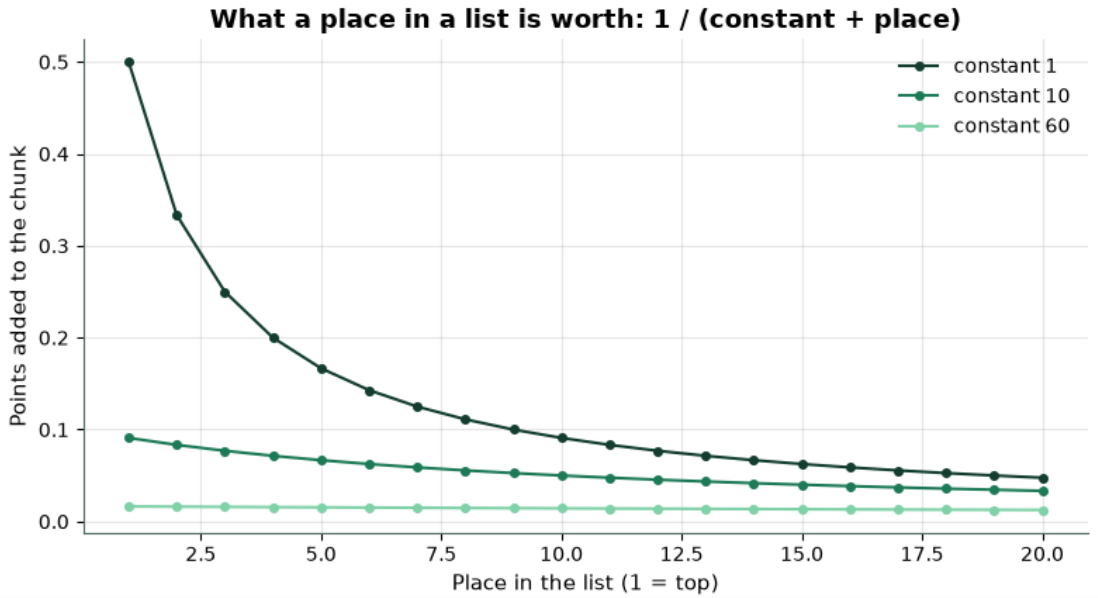
A chunk missing from a list adds nothing

c = 60; ties by chunk id

By hand: a b c and b c d

```
place 1: b  0.032522
place 2: c  0.032002
place 3: a  0.016393
place 4: d  0.015873
```

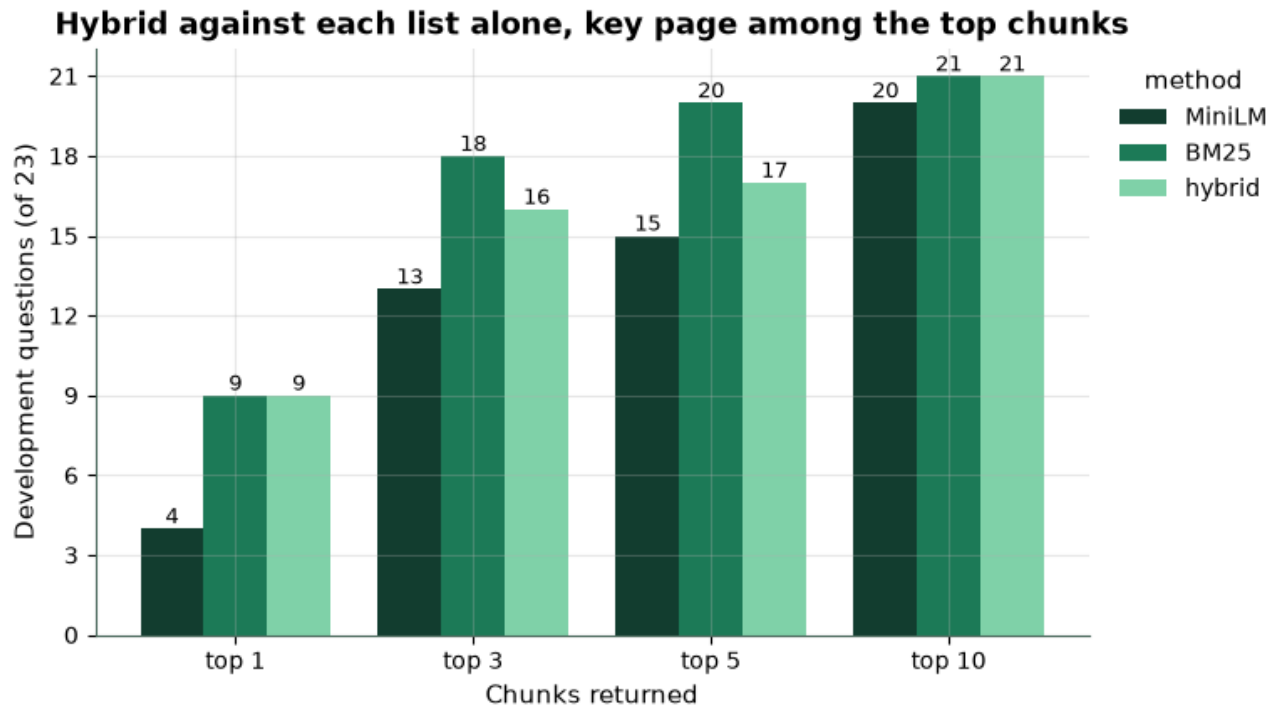
dev21: 34 chunks in the two top-20 lists; first on the key's page: fused 1, MiniLM 3, BM25 6.



Cormack, Clarke, Buettcher (SIGIR 2009)

Reciprocal rank fusion outperforms condorcet and individual rank learning methods, pp. 758-759, published 19 July 2009. Crossref's record of DOI 10.1145/1571941.1572114, read 13 October 2026. The paper itself was not read.

Hybrid against each list alone



Key page, 23 questions

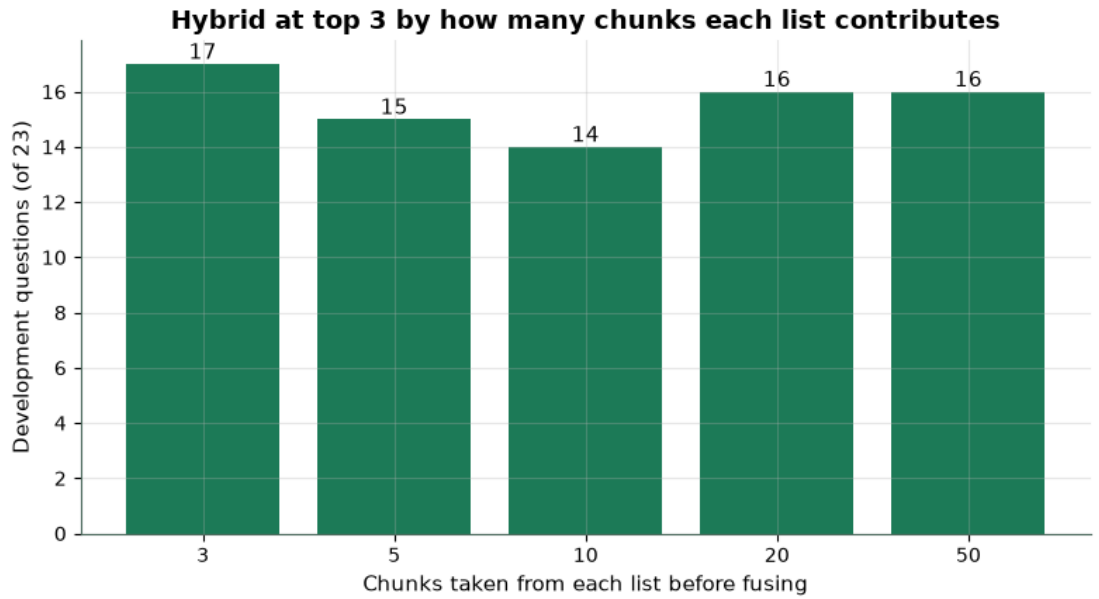
method	top 1	top 3	top 5	top 10
MiniLM	4	13	15	20
BM25	9	18	20	21
hybrid	9	16	17	21

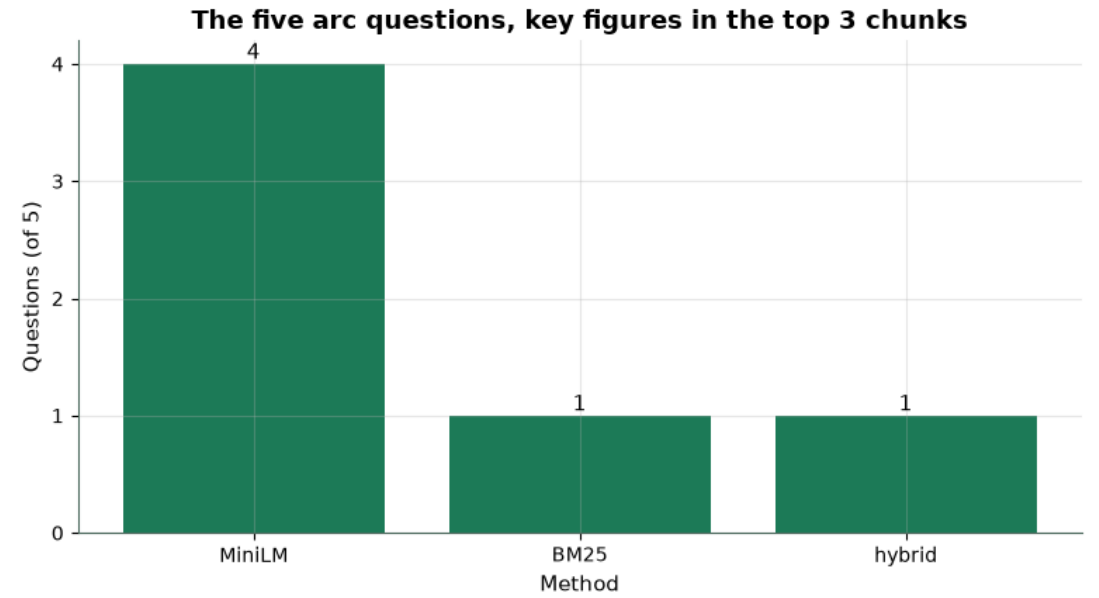
against BM25 at top 3: gains dev21

loses dev07, dev08, dev13

Fusion combines lists; it has no guarantee of beating either. A finding about these questions and these lists.

Depth of the lists, and the arc questions





Hybrid hits at top 3 by depth: 3 -> 17, 5 -> 15, 10 -> 14, 20 -> 16, 50 -> 16

Five arc questions, key figures in the top 3 chunks: MiniLM 4, BM25 1, hybrid 1

Which method has more hits depends on the questions asked; five questions is a very small set.

Excel side by side, and Git

Excel side by side (tested in Excel)

`=LN((E2-D2+0.5)/(D2+0.5)+1)`	idf; equal to Python on 9 rows
`=J2*C2*(H2+1)/(C2+H2*(1-I2+I2*F2/G2))`	weight 0.1245 for text A, senior; equal on 9 rows
`=SUMIFS($K$2:$K$10,$A$2:$A$10,"A")`	A 1.1634, B 0.2393, C 0.4410
`one weight in the 10-K: sofr in boyd10k-8-101-1072`	6.4526 = Python (N 1158, n 17, f 3, dl 88)
`=1/(60+B2), then =SUMIFS($D$2:$D$41,$C$2:$C$41,F2)`	fused scores equal rrf for 34 chunks (1e-15)
`=RANK(G2,$G$2:$G$35)`	ties share a place; rrf breaks ties by id: 9 chunks differ

Git: rag_results.csv with 3 retrieval rows, then a fourth run and its diff; 38 tests.

The AI check

The docstring is the prompt. The retrieval code that comes back runs and looks finished.

1

Docstring

Units, conventions, and what the code must never let through: what you asked for.

→

2

Code back

Written for the course in the style of an AI answer. Read it.

→

3

Test first

Plant the case the docstring promises to catch. Run it: it fails.

→

4

Fix

Find the rule it broke, fix it, run the test again.

Today: the length is counted in characters, divided by an average in tokens, so every length penalty is too large.

Code that runs is not the same as code that is right.

What you can do now

- Search by meaning and by keyword differ on exact strings
- Work BM25 by hand; index with bm25_index, tokens as length
- Count hits by a named rule, on questions set aside
- Fuse two lists with rrf; hybrid has no guarantee
- Which method has more hits depends on the questions

Week 2, Day 2 Reranking with a cross-encoder